

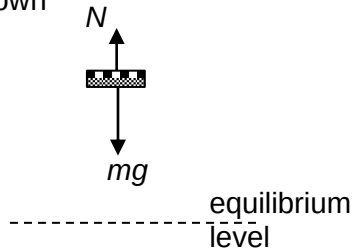
- 13 A When the platform is above the equilibrium level, and moving down

$$mg - N = m\omega^2 x$$

When the coin loses contact with the platform, $N = 0$

$$\therefore mg = m\omega^2 x_0 \Rightarrow \omega^2 x_0 = g$$

$$\Rightarrow x_0 = \frac{g}{\omega^2} = \frac{9.81}{(2\pi \times 3.2)^2} = 0.0243 \text{ m}$$



- 14 A This is derived from $nV = \frac{1}{m} N m c^2$ and is not an assumption